



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS ADVANCED

2019 Year 11 Course Assessment Task 2

y-day, xx May 2019

General instructions

- Working time – 1 hour.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 3)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Marker's use only.

QUESTION	1-5	1	2	3	??	??	Total	%
MARKS	$\overline{5}$	$\overline{10}$	$\overline{6}$	$\overline{4}$	$\overline{11}$	$\overline{7}$	$\overline{43}$	

Section I

5 marks

Attempt Question 1 to 5

Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided (labelled as page 3).

Questions

Marks

1. What is the solution to the equation $\cos\left(\frac{\theta}{2} + 30^\circ\right) = \sin \theta$ for $0^\circ \leq \theta \leq 90^\circ$? 1

(A) $\theta = 20^\circ$ (B) $\theta = 30^\circ$ (C) $\theta = 40^\circ$ (D) $\theta = 50^\circ$

2. What are the values of a and b if $a + b\sqrt{2} \equiv \frac{2}{3 + \sqrt{2}}$? 1

(A) $a = \frac{6}{7}, b = -\frac{2}{7}$ (C) $a = \frac{6}{5}, b = \frac{2}{5}$

(B) $a = \frac{6}{7}, b = \frac{2}{7}$ (D) $a = \frac{6}{5}, b = -\frac{2}{5}$

3. Determine the type of roots for the following equation:

$$4x^2 - 4x + 1 = 0$$

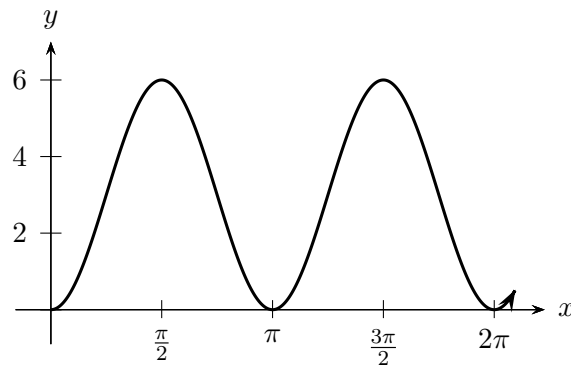
(A) Two distinct, real roots (C) One pair of unreal roots
(B) Irrational roots (D) One pair of equal roots

4. What is $\frac{\tan \theta \sec \theta}{1 + \tan^2 \theta}$ in simplest form?

(A) $\sin \theta$ (B) $\cos \theta$ (C) $\tan \theta$ (D) $\sin \theta \cos \theta$

5. The equation of the graph below takes the form $y = A \cos Bx + 3$.

1



Which of the following are the values of A and B ?

(A) $A = 3, B = -2$

(C) $A = 6, B = 3$

(B) $A = -3, B = 2$

(D) $A = -6, B = \pi$

Answer grid for Questions 1 - 5

- | | | | | | |
|---|---|-----|-----|-----|-----|
| 1 | — | (A) | (B) | (C) | (D) |
| 2 | — | (A) | (B) | (C) | (D) |
| 3 | — | (A) | (B) | (C) | (D) |
| 4 | — | (A) | (B) | (C) | (D) |
| 5 | — | (A) | (B) | (C) | (D) |

Examination continues overleaf...

Question 1 (8 Marks) Commence a NEW booklet. **Marks**

(a) Fully simplify:

i. $8x - 2y + 4x + 9x + z - 2y.$ **1**

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ii. $(2p - 5)^2 - (2p + 5)^2.$ **2**

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(b) $\frac{5}{2k+6} + \frac{k}{k^2-9}.$ **3**

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(c) Solve for x and y : **2**

$$\begin{cases} 7x - 3y = 41 \\ 3x - y = 17 \end{cases}$$

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Question 2 (22 Marks)

Commence a NEW booklet.

Marks

- (a) Is the function $f(x) = x^3 - x$ odd or even? Justify your answer with the appropriate reasoning. **2**

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- (b) Find the domain and range of the following, writing in interval notation:

- i. $y = 3x^2 - 1$. **1**

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- ii. $y = \frac{1}{x-2}$. **2**

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- iii. $y = -\sqrt{9-x^2}$. **2**

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- (c) Suppose $f : x \mapsto x^2 + 1$ and $g : x \mapsto 3 - x$.

Find:

- i. $f \circ g(2)$. **1**

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- ii. $g \circ f(x)$, in simplest form. **2**

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- iii. Find the value(s) of x such that $g \circ f(x) = f(x)$. **2**

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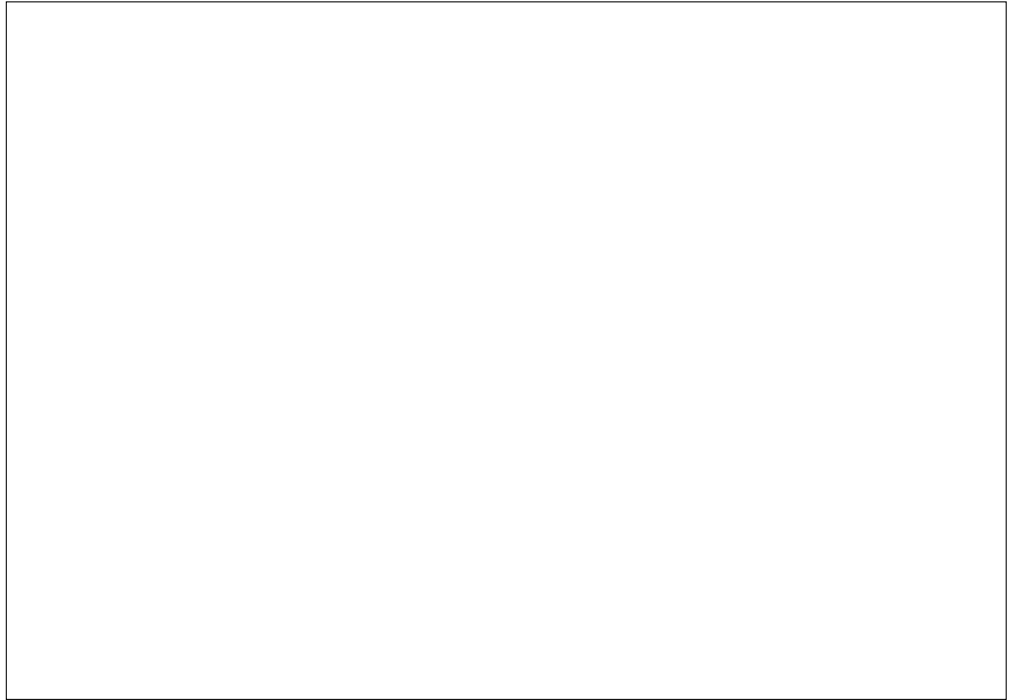
- (d) Solve for x : $|7 - 3x| = 1$. **2**

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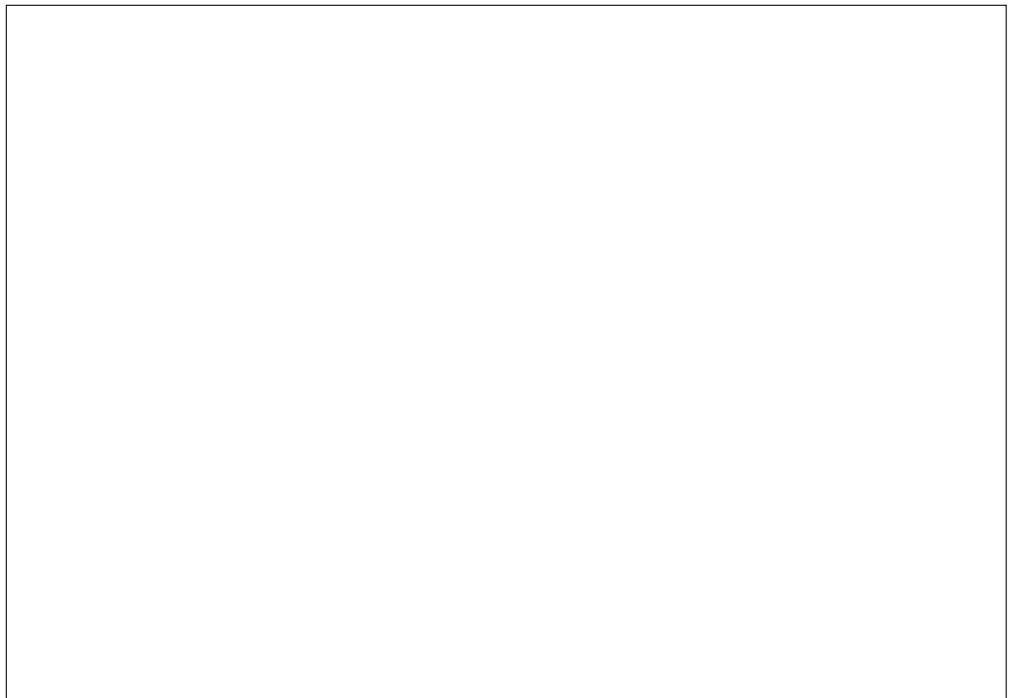
Question 2 continues overleaf...

- (e) Sketch the following graphs on separate number planes. Showing all important features, including x and y intercepts.

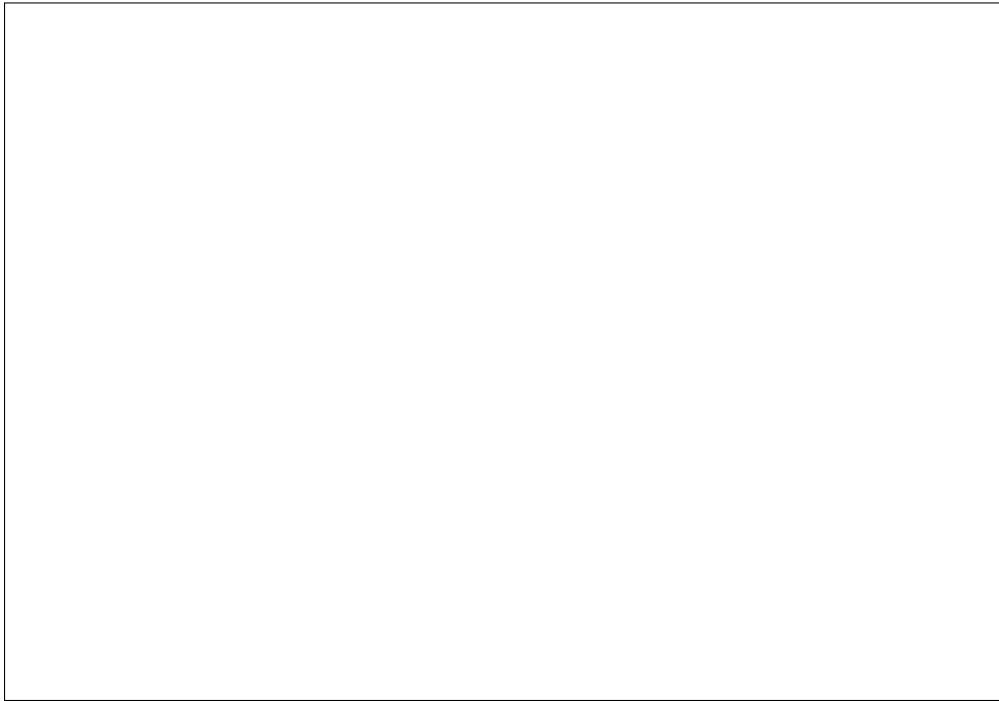
i. $y = (x - 1)(x + 3)(4 - x)$.

2

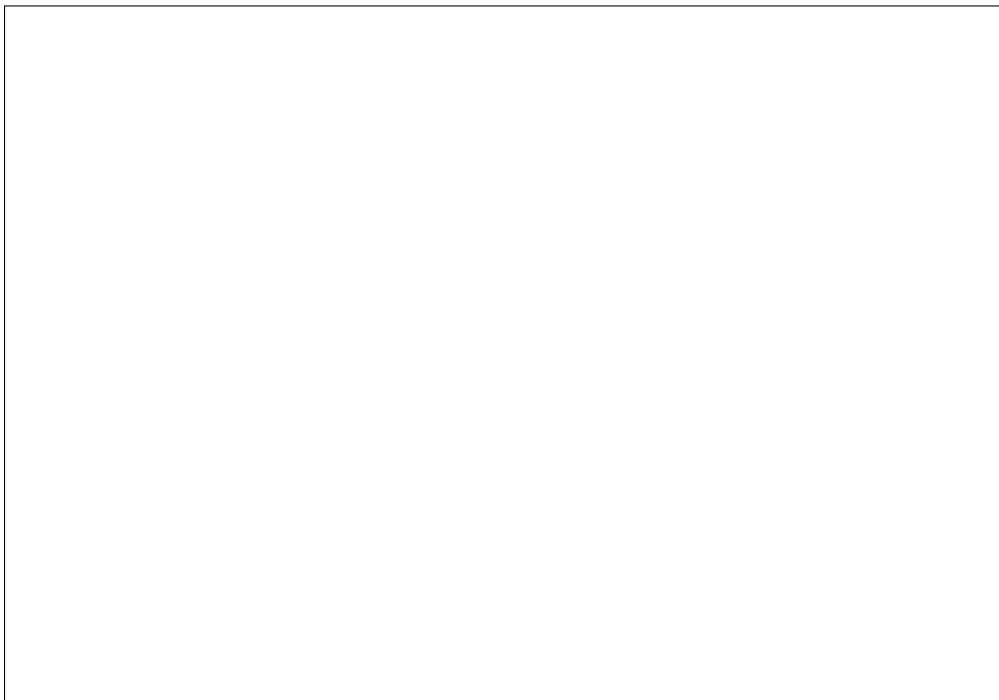
ii. $y = -x^2 - 4x + 5$.

2

iii. $y = \sqrt{x+3}$

2

iv. $y = |x-2| + 1$

2

Question 3 (17 Marks)

Commence a NEW booklet.

Marks

- (a) In $\triangle ABC$, $\angle BAC = 50^\circ$, $AC = 40$ cm and $BC = 25$ cm.

3

Find the size of $\angle ABC$ correct to the nearest minute.

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- (b) Solve for θ :

$$4\sin^2\theta - 3 = 0 \quad 0^\circ \leq \theta \leq 360^\circ$$

3

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- (c) If $\sin x = -\frac{2}{5}$ and $\cos x > 0$, find the exact value of $\tan x$.

2

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- (d) Fully simplify:

3

$$\frac{\cos \theta - 2 \cos^3 \theta}{\sin \theta - 2 \sin^3 \theta}$$

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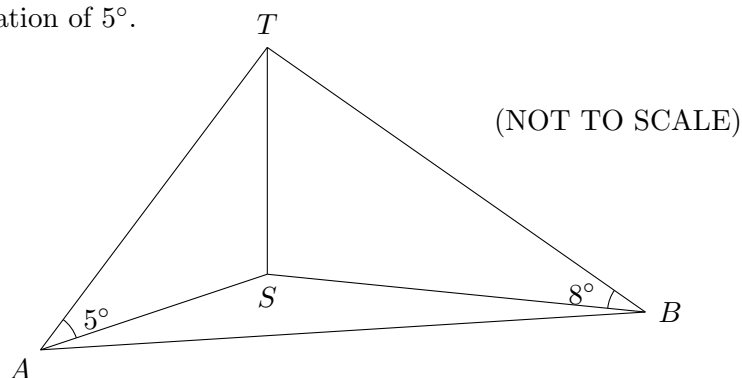
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- (e) A man walking along a straight road observes a spire bearing 045°T with an angle of elevation of 5° .



After travelling a distance of 5 km, the spire bears 315°T and the angle of elevation is 8° .

- i. Find the size of $\angle ASB$.

1

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- ii. Show that $AS = h \tan 85^\circ$, and write a similar expression for BS .

2

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- iii. Find the height of the spire, correct to the nearest metre.

3

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End of paper.

2019 Mathematics Advanced (Year 11 Course) Assessment Task 2 STUDENT SELF REFLECTION

1. In hindsight, did I do the best I can? Why or why not?

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2. Which topics did I need more help with, and what parts specifically?

- Q1, 2 - Functions, absolute value inequality

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- Q3 - Trigonometric Ratios

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- Q4 - Trigonometric Ratios (Graphs, identities)

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- Q5 - Plane Geometry

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- Q6 - Applied Trigonometry

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- Q7 - Problem solving

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3. What other parts from the teacher feedback can I take away to refine my solutions for future reference?

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SOLUTIONS

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- Attempt all questions.

SECTION I

- Mark your answers on the answer grid provided (on page 3)

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- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

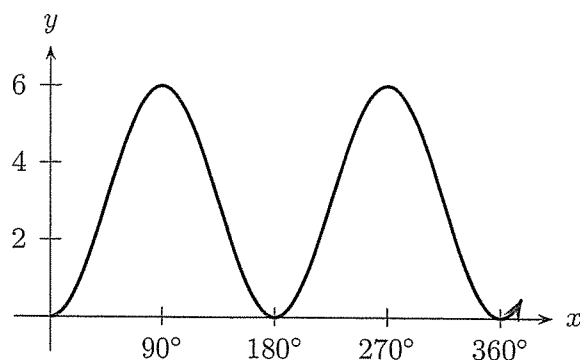
NESA STUDENT #: # BOOKLETS USED:

Marker's use only.

QUESTION	1-5	6	7	8	Total	%
MARKS	$\overline{5}$	$\overline{8}$	$\overline{22}$	$\overline{16}$	$\overline{51}$	

5. The equation of the graph below takes the form $y = A \cos Bx + 3$.

1



Which of the following are the values of A and B ?

(A) $A = 3, B = -2$

(C) $A = 6, B = 3$

(B) $A = -3, B = 2$

(D) $A = -6, B = 180^\circ$

Answer grid for Questions 1 - 5

1 -	☐ (A)	☐ (B)	☒ (C)	☐ (D)
2 -	☒ (A)	☐ (B)	☐ (C)	☐ (D)
3 -	☐ (A)	☐ (B)	☐ (C)	☒ (D)
4 -	☒ (A)	☐ (B)	☐ (C)	☐ (D)
5 -	☐ (A)	☒ (B)	☐ (C)	☐ (D)

Examination continues overleaf...

Section II

46 marks

Attempt Question 6 to 8

Allow approximately 50 minutes for this section

Write your answers in the space provided.

Question 6 (8 Marks)

Marks

(a) Fully simplify:

i. $8x - 2y + 4z + 9x + z - 2y.$

1

$$= 17x - 4y + 5z \quad \checkmark$$

ii. $(2p - 5)^2 - (2p + 5)^2.$

2

$$= [2p - 5 + 2p + 5][2p - 5 - 2p - 5] \quad \checkmark$$

$$= 4p(-10)$$

$$= -40p \quad \checkmark$$

iii. $\frac{5}{2k+6} + \frac{k}{k^2-9}.$

3

$$= \frac{5}{2(k+3)} + \frac{k}{(k+3)(k-3)} \quad \checkmark$$

$$= \frac{5(k-3) + 2k}{2(k+3)(k-3)} \quad \checkmark$$

$$= \frac{7k-15}{2(k+3)(k-3)} \quad \checkmark$$

(b) Solve for x and y :

2

$$\begin{cases} 7x - 3y = 41 & \text{--- (1)} \\ 3x - y = 17 & \text{--- (2)} \end{cases}$$

$$\textcircled{2} \times 3,$$

$$9x - 3y = 51 \text{ --- } \textcircled{3}$$

$$\textcircled{3} - \textcircled{1},$$

$$2x = 10$$

$$x = 5$$

✓

Sub into $\textcircled{2}$,

$$15 - y = 17$$

$$y = -2$$

✓

$$\therefore x = 5, y = -2$$

Question 7 (22 Marks)

Marks

- (a) Is the function $f(x) = x^3 - x$ odd or even? Justify your answer with the appropriate reasoning. 2

$$\begin{aligned}
 f(x) &= x^3 - x & \therefore f(x) \text{ is odd } \checkmark \\
 f(-x) &= (-x)^3 - (-x) \\
 &= -x^3 + x \\
 &= -f(x) \quad \checkmark
 \end{aligned}$$

- (b) Find the domain and range of the following, writing in interval notation:

i. $y = 3x^2 - 1$.

2

$$\begin{aligned}
 D &: \{x : x \in \mathbb{R}\} \quad \checkmark \\
 R &: \{y : y \in [-1, \infty)\} \quad \checkmark
 \end{aligned}$$

ii. $y = \frac{1}{x-2}$.

2

$$\begin{aligned}
 D &: \{x : x \in \mathbb{R} \setminus 2\} \quad \checkmark \\
 R &: \{y : y \in \mathbb{R} \setminus 0\} \quad \checkmark
 \end{aligned}$$

iii. $y = -\sqrt{9-x^2}$.

2

$$\begin{aligned}
 D &: \{x : x \in [-3, 3]\} \quad \checkmark \\
 R &: \{y : y \in [-3, 0]\} \quad \checkmark
 \end{aligned}$$

Q7(b) Deduct 1 mark if answers not written in interval notation.

- (c) Suppose $f : x \mapsto x^2 + 1$ and $g : x \mapsto 3 - x$.

$$f(x) = x^2 + 1$$

$$g(x) = 3 - x$$

Find:

i. $(f \circ g)(2) = f(g(2))$

1

$$= f(1)$$

$$= 2$$

ii. $(g \circ f)(x)$, in simplest form.

$$= g(f(x))$$

$$= g(x^2 + 1)$$

$$= 3 - (x^2 + 1)$$

$$= 2 - x^2$$

iii. Find the value(s) of x such that $(g \circ f)(x) = f(x)$.

2

$$g(f(x)) = f(x)$$

$$2 - x^2 = x^2 + 1$$

$$2x^2 = 1$$

$$x = \pm \frac{1}{\sqrt{2}}$$

(d) Solve for x : $|7 - 3x| = 1$.

2

$$7 - 3x = 1$$

or

$$7 - 3x = -1$$

$$3x = 6$$

$$3x = 8$$

$$x = 2$$

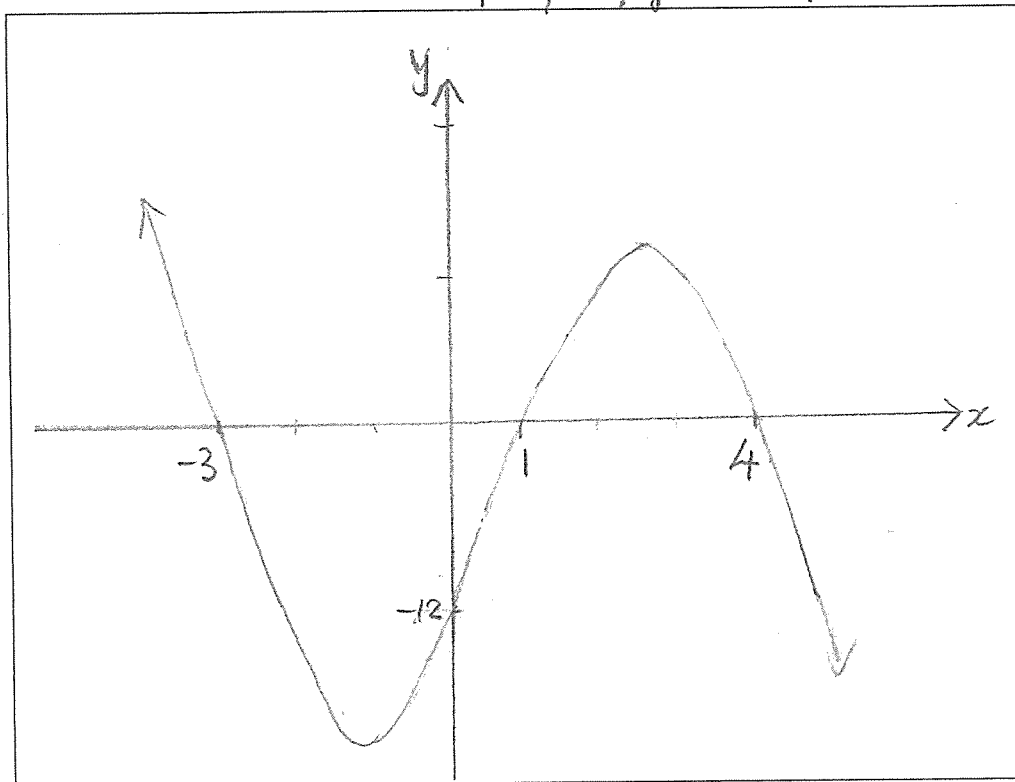
$$x = \frac{8}{3}$$

Question 7 continues overleaf...

- (e) Sketch the following graphs on separate number planes. Showing all important features, including x and y intercepts.

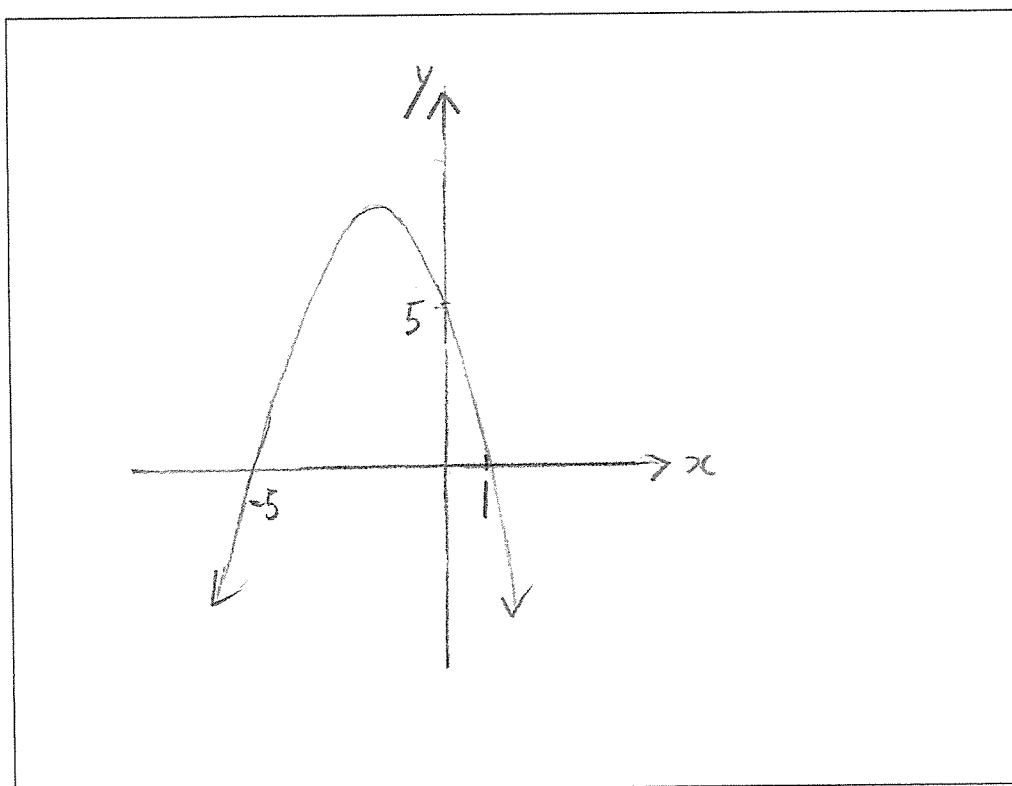
i. $y = (x - 1)(x + 3)(4 - x)$.

shape ✓
Critical point / x / y intercepts labelled ✓ 2



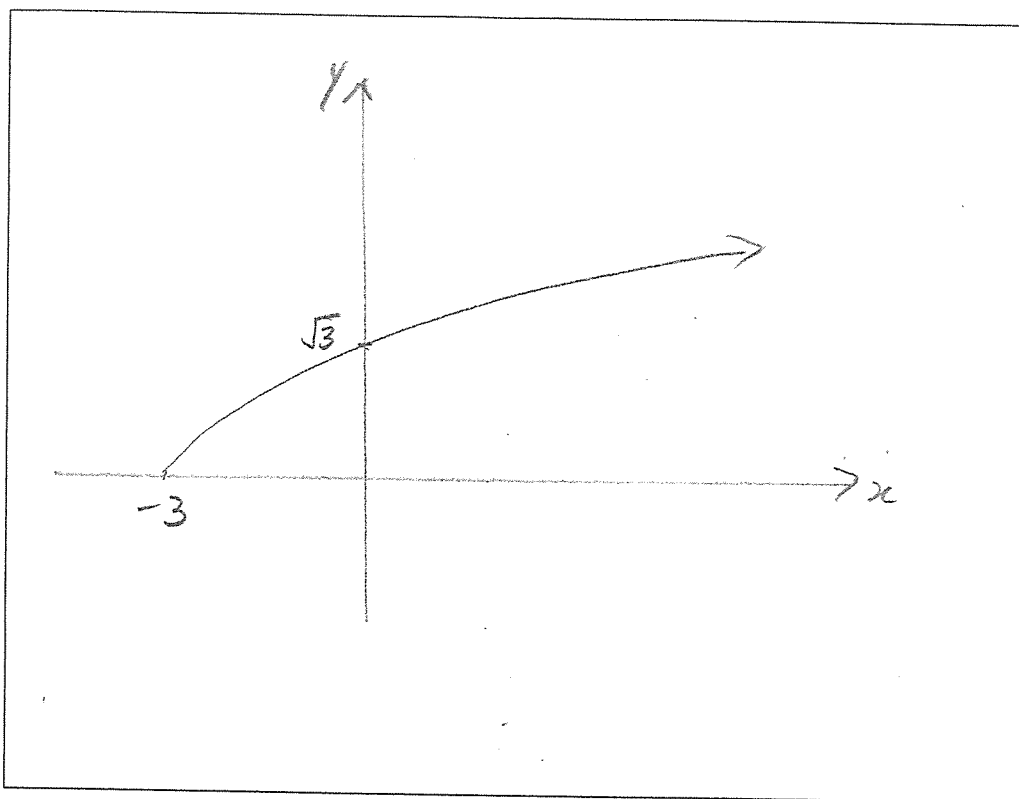
ii. $y = -x^2 - 4x + 5 = (x+5)(1-x)$

2



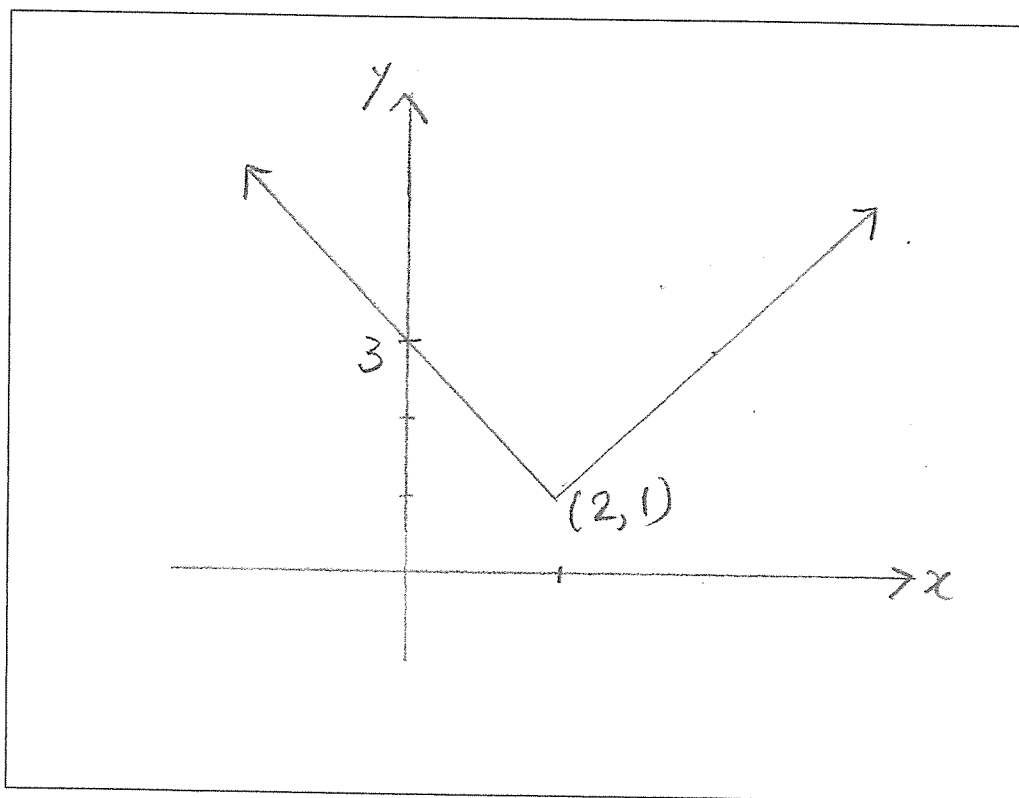
iii. $y = \sqrt{x+3}$

2



iv. $y = |x-2|+1$

2



Question 8 (16 Marks)

Marks

- (a) In $\triangle ABC$, $\angle BAC = 50^\circ$, $AC = 30$ cm and $BC = 25$ cm.

3

Find the size of $\angle ABC$ correct to the nearest minute.

$$\frac{30}{\sin \angle ABC} = \frac{25}{\sin 50^\circ} \quad \checkmark$$

$$\sin \angle ABC = \frac{30 \sin 50^\circ}{25}$$

$$= 0.91925 \dots$$

$$\angle ABC = 66^\circ 49' \text{ or } 113^\circ 11' \quad \checkmark$$

- (b) Solve for θ :

2

$$4 \sin^2 \theta - 3 = 0 \quad 0^\circ \leq \theta \leq 360^\circ$$

$$\sin^2 \theta = \frac{3}{4}$$

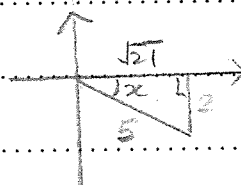
$$\sin \theta = \pm \frac{\sqrt{3}}{2} \quad \checkmark$$

$$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ \quad \checkmark$$

- (c) If $\sin x = -\frac{2}{5}$ and $\cos x > 0$, find the exact value of $\tan x$.

2

$$\tan x = -\frac{2}{\sqrt{21}} \quad \checkmark$$

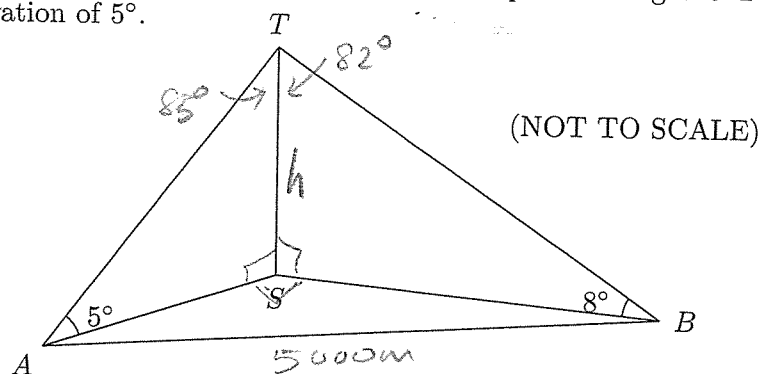


(d) Fully simplify:

3

$$\frac{\cos \theta - 2 \cos^3 \theta}{\sin \theta - 2 \sin^3 \theta}$$

$$\begin{aligned}
 &= \frac{\cos \theta (1 - 2 \cos^2 \theta)}{\sin \theta (1 - 2 \sin^2 \theta)} \checkmark \\
 &= \frac{\cos \theta [1 - 2(1 - \sin^2 \theta)]}{\sin \theta (1 - 2 \sin^2 \theta)} \checkmark \\
 &= \frac{\cos \theta (-1 + 2 \sin^2 \theta)}{1 - 2 \sin^2 \theta} \\
 &= -\cos \theta \checkmark
 \end{aligned}$$

(e) A man walking along a straight road observes a spire bearing 045°T with an angle of elevation of 5° .After travelling a distance of 5 km, the spire bears 315°T and the angle of elevation is 8° .i. Find the size of $\angle ASB$.

$$\begin{aligned}
 \angle ASB &= 45^\circ + 45^\circ \\
 &= 90^\circ \checkmark
 \end{aligned}$$

ii. Show that $AS = h \tan 85^\circ$ where h is the height of the spire in metres. Write a similar expression for BS .

$$\text{In } \triangle AST, \angle ATS = 90^\circ - 5^\circ = 85^\circ$$

$$\tan 85^\circ = \frac{AS}{h}$$

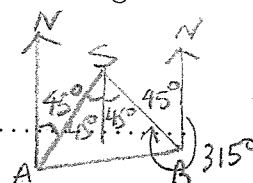
$$\therefore AS = h \tan 85^\circ \checkmark$$

In $\triangle BST$,

$$\begin{aligned}
 \angle STB &= 90^\circ - 8^\circ \\
 &= 82^\circ
 \end{aligned}$$

$$\tan 82^\circ = \frac{BS}{h}$$

$$\therefore BS = h \tan 82^\circ \checkmark$$



- iii. Find the height of the spire, correct to the nearest metre.

3

In $\triangle ABC$

$$AS^2 + BS^2 = AB^2$$

$$h^2 \tan^2 85^\circ + h^2 \tan^2 82^\circ = 5000^2 \quad \checkmark$$

$$h^2 (\tan^2 85^\circ + \tan^2 82^\circ) = 5000^2$$

$$h = \frac{5000}{(\tan^2 85^\circ + \tan^2 82^\circ)^{\frac{1}{2}}} \quad \checkmark$$

$$= 371 \text{ m (nearest m)} \quad \checkmark$$

End of paper.